

PART I

Introduction to the serial communication port of V60

1. Introduction

V60 is equipped with two serial communication port, one is RS485, and one is IR

2. Default settings of serial communication port

RS485,logic level USART 9600,N,8,1

IR 2400,N,8,1

3. How to Change the serial ports settings

The changes of the serial port settings can only be realized by using the V60_SETUP, which is a PC based software.

The serial port settings are displayed on M0E display window

PART II

The MODBUS protocol Application

1. MODBUS default setup by the manufacturer

1.1 RS485

Baud Rate	9600	(possible setting	300,600,1200,2400,4800,9600)
Parity	None	(possible setting	None, Even, Odd)
Address	1		

1.2 IR

Baud Rate	9600	(possible setting	300,600,1200,2400,4800,9600)
Parity	None	(possible setting	None, Even, Odd)
Address	1		

All settings can be displayed on M0E, and M10

2. MOSBUS Misc about V60

2.1

2.2

2.3

2.4

3. MODBUS Table of Common Holding Register Addresses

REGISTER	NUMBER	VARIABLE NAME	FORMAT	NOTE
0001-0002	2	Flow Rate	IEEE754	Unit: m3/h
0003-0004	2	Energy Flow Rate	IEEE754	Unit: kW
0005-0006	2	Velocity of the flow	IEEE754	Unit: m/s
0009-0010	2	Positive Flow accumulator Note 1	LONG	Unit:m3,L,GAL,ft3
0011-0012	2	Positive flow decimal fraction	IEEE754	
0013-0014	2	Negative flow accumulator	LONG	Unit:m3,L,GAL,ft3
0015-0016	2	Negative flow decimal fraction	IEEE754	Singular real number, also called FLOAT
0017-0018	2	Positive energy accumulator	LONG	Unit: kWh, GJ, KBTU
0019-0020	2	Positive energy decimal fraction	IEEE754	
0021-0022	2	Negative energy accumulator	LONG	Unit: kWh, GJ, KBTU
0023-0024	2	Negative energy decimal fraction	IEEE754	
0025-0026	2	Net flow accumulator	LONG	Unit:m3,L,GAL,ft3
0027-0028	2	Net flow decimal fraction	IEEE754	
0029-0030	2	Net energy accumulator	LONG	Unit: kWh, GJ, KBTU
0031-0032	2	Net energy decimal fraction	IEEE754	
0033-0034	2	Temperature T1 (Supply)	IEEE754	Unit: °C
0035-0036	2	Temperature T2 (Return)	IEEE754	Unit: °C
0053-0055	3	Calendar (date and time) 6 Bytes of BCD stands SMHDMY, lower byte first	BCD	Writable.
0056	1	Day+Hour for Auto-Save For example 0512H stands Auto-save on 12:00 on 5 th . 0012H for 12:00 on everyday.	BCD	Writable.
0057	1	Password for writing	Integer	Writable
0058	1	Password to enter Sleep mode	Integer	Writable., Write 0x5A58, to enter sleep mode
0059	1	Key Input	Integer	Writable
0060	1	Go to Window #	Integer	Writable.

0061	1	Current display window	Integer	Writable
0062	1	Primary communication Address.	Integer	Writable. Max 255
0063	1	BC timer	Integer	Write 0 to start BC
0064	1	OCT1 pulse	Integer	
0065	1	OCT2 pulse	Integer	
0071	1	Supplementary ErrCode	Bits	Note 4
0072	1	ErrCode	Bits	Note 5
0077-0078	2	Resistance of the T1 sensor	IEEE754	Ω
0079-0080	2	Resistance of the T2 sensor	IEEE754	Ω
0081-0082	2	Total travel time	IEEE754	Unit: uS
0083-0084	2	Delta travel time	IEEE754	Unit: nS
0092	1	Signal Quality	Integer	Channel 1 in Low byte
0093	1	Signal Strength of Channel #1	Integer	Range:0~4095
0094	1	Signal Strength of Channel #2	Integer	
0095	1	Battery voltage r	Integer	$V = \text{REG95} * (2.5/4096)$
0099-0100	2	Reynolds' Number	IEEE754	
0101-0102	2	Correction for Reynolds' number	IEEE754	
0105-0106		Total working hours	Long	Unit: Seconds
0107-0108		Power on times counter	Long	
0109-0110		CPU temperature	IEEE754	Unit: $^{\circ}\text{C}$
0113-0114	2	NET Total flow (IEEE754)	IEEE754	Unit: m3/h
0115-0116	2	POS Total flow (IEEE754)	IEEE754	Signal Accuracy.
0117-0118	2	NEG Total flow (IEEE754)	IEEE754	
0119-0120	2	NET Total energy (IEEE754)	IEEE754	Not Recommended to read these Registers with new developments due to limited accuracy
0121-0122	2	POS Total energy (IEEE754)	IEEE754	
0123-0124	2	NEG Total energy (IEEE754)	IEEE754	
0125-0126	2	NET Total flow today (IEEE754)	IEEE754	
0127-0128	2	NET month total flow (IEEE754)	IEEE754	
0129-0130	2	Tariff 2 accumulator	LONG	
0131-0132	2	Tariff 2 accumulator decimal fraction	IEEE754	
0133-0134	2	Tariff 3 accumulator	LONG	
0135-0136	2	Tariff 3 accumulator decimal fraction	IEEE754	
0137-0138	2	Today's Flow accumulator	LONG	9 Digits
0139-0140	2	Today's flow decimal fraction	IEEE754	
0141-0142	2	This month's Flow accumulator	LONG	
0143-0144	2	This month's flow decimal fraction	IEEE754	
0144-0145	2	This year's Flow accumulator	LONG	
0147-0148	2	This year's flow decimal fraction	IEEE754	
0149-0150	2	Today's Energy accumulator	LONG	
0151-0152	2	Today's Energy decimal fraction	IEEE754	
0153-0154	2	This month's Energy accumulator	LONG	
0155-0156	2	This month's Energy decimal fraction	IEEE754	
0162	1	Daily data pointer	Integer	
0163	1	Month data pointer	Integer	
0165-0166		Failure Timer	Long	Unit: Seconds
0181-0182	2	Temperature difference	IEEE754	Unit: $^{\circ}\text{C}$
0185-0186		Main Clock Difference	IEEE754	
0187-0188		Auto-Save: total time	Long	
0189-0190		Auto-Save: POS total flow	Long	

0191-0192		Auto-Save: Flow Rate	IEEE754	
0193-0194		Auto-Save: total NEG timer	Long	
0195-0196		Auto-Save: NEG total flow	Long	
0197-0198	2	Calibrating flow totalizer	Long	Unit:m3,GAL,ft3 , L
0199-0200	2	Calibrating flow decimal fraction	IEEE754	
0201-0202	2	Calibrating energy totalizer	long	Unit: kWh, GJ, KBTU
0203-0204	2	Calibrating energy decimal fraction	IEEE754	
0205	1	Calibrating Timer	integer	Unit in 250mS
0221-0222	2	DN of the pipe	IEEE754	Unit:mm
0259-0260	2	Max flow rate of the month	IEEE754	Unit: m3/h
0261-0262	2	Max heat rate of the month	IEEE754	Unit: kW
0263-0264	2	Max inlet temperature of the month	IEEE754	
0265-0266	2	Max Outlet temperature of the month	IEEE754	
0267-0268	2			
0269	1			
0270	1			
0271-0272	2	Delta Tau	IEEE754	In unit nS
0273-0274	2	M-bus secondary address	BCD	
0275-0276	2	Negative flow timer	long	In unit seconds
0277-0280	4			
0281-0282	2			
0283-0284	2			
0285-0286	2	Max flow rate of the day	IEEE754	Unit: m3/h
0287-0288	2	Max energy rate of the day	IEEE754	Unit: kW
0289-0290	2	Max inlet temperature of the day	IEEE754	Unit: °C
0291-0292	2	Max outlet temperature of the day	IEEE754	Unit: °C
0293-0294	2			
0295-0296	2	CUSTOMER'S NUMBER	BCD	
0297-0298	2	Time set point to stop tariff total	BCD	
0299-0300	2	Time set point to start tariff2 total	BCD	
0301-0302	2	Time set point to start tariff3 total	BCD	
0303	0.5	State byte for tariff total and batch	BCD	Low byte
0303-0304	1.5	#1 time set point to trig batch	BCD	
0305-0306	1.5	#2 time set point to trig batch	BCD	
0306-0307	1.5	#3 time set point to trig batch	BCD	
0307-0308	1.5	#5 time set point to trig batch	BCD	
0309-0310	1.5	#5 time set point to trig batch	BCD	
0311-0312	2	Set point for batch controller	IEEE754	
0359	1	Simulation Mode Control Word	Integer	Valid with V60.05+
0361-0362	2	Always read as 361.00 Note 2	IEEE754	For test purpose
0363-0364	2	Always read as 363348858	long	
0365-0366	2	Always read as -987654321	long	
1438	1	Total flow unit number	INTEGER	0=cubit meter 1=liter 2=GAL 5=cubic feet
1439	1	Total flow multiplier	Integer	n:(-4..3), Note1
1440	1	Total Energy multiplier	Integer	n:(-3..4), Note1
1441	1	Total energy unit number	Integer	0=GJ , 2=kWH 1=Kilo BTU
1491	1	Meter Type	Integer	EN1434-3
1527	1	Version of Software Note3	Integer	

1528	1	Manufacturer	Note3	Integer	Value=0x1188
1529	2	ESN		BCD	MSB first

Note 1 (1) The internal accumulator is been presented by a LONG number for the integer part together with a REAL number for the decimal fraction. In general uses, only the integer part needs to be read. Reading the fraction can be omitted. The final accumulator result has a relation with unit and multiplier. Assume N stands for the integer part (for the positive accumulator, the integer part is the content of REG 0009, 0010, a 32-bits signed LONG integer,), Nf stands for the decimal fraction part (for the positive accumulator, the fraction part is the content of REG 0011, 0012, a 32-bits REAL float number,), n stands for the flow decimal point (REG 1439).

then

The final positive flow rate= $(N+Nf) \times 10^{n-3}$ (in unit decided by REG 1439)

The meaning of REG 1438 which has a range of 0~3 is as following:

- | | | |
|---|-----------------|-------|
| 0 | cubic meter | (m3) |
| 1 | liter | (L) |
| 2 | American gallon | (GAL) |
| 3 | Cubic feet | (CF) |

For example, if REG0009=123456789, REG0010=0.123, and REG1439=-1, REG1438=0

Then the positive flow is 12345.6789123 m3

For energy totlizers

The energy flow rate $= (N+Nf) \times 10^{n-4}$

n was decided by REG01440

The energy unit decided by REG 1441).

Note 2 Correctly reading REG361 as a float number shall always be 361.00. If a “0” or “250.264” is read by REG ”361”, then it means the used address is wrong.

Note3 The REGs for version and manufacturer is used to identify this MODBUS register table.

4. MODBUS Table of Monthly Totalizer's Addresses

Data Blk No	address		Reg no	Variable name	format	note
n/a	0163		1	Monthly Pointer	Integer	Range: 0-127
0	0	29953	1	Data block no	Integer	0~65535
	1	29954	1	State	Integer	
	2	29955	1	Failure days	Integer	
	3	29956	1	Month, Year	BCD	Month in low byte
	4	29957	2	Working hours	LONG	Record for checking
	6	29959	2	Failure hours	LONG	
	8	29961	2	Month Total flow	LONG	Of this month
	10	29963	2	Month Total energy	LONG	Of this month
	12	29965	2	Positive flow	LONG	Recorded
	14	29967	2	Negative flow	LONG	Recorded
	16	29969	2	Positive energy	LONG	Recorded
	18	29971	2	Negative energy	LONG	Recorded
	20	29973	2	Tariff2	LONG	Recorded
	22	29975	2	Tariff3	LONG	Recorded
	24	29977	2	Max flow rate	REAL4	
	26	29979	2	Max energy rate	REAL4	
	28	29981	2	Max inlet temperature	REAL4	
	30	29983	2	Max outlet temperature	REAL4	
2	0	29985	1	Data block no	Integer	
	1	29986	1	State	Integer	
	2	29987	1	Failure days	Integer	
	3	29988	1	Month, Year	BCD	
	4	29989	2	Working hours	LONG	
	6	29991	2	Failure hours	LONG	
	8	29993	2	Month Total flow	LONG	
	10	29995	2	Month Total energy	LONG	
	...					
	28	30013	2	Max inlet temperature	REAL4	
	30	30015	2	Max outlet temperature	REAL4	
n	Monthly data block n					
127	0	38113	1	Data block no	Integer	
	1	38114	1	State	Integer	
	...					
	28	38141	2	Max inlet temperature	REAL4	
	30	38143	2	Max outlet temperature	REAL4	

5. MODBUS Table of Daily Totlizer's Addresses

Data Blk No	address		Reg no	Variable name	format	note
n/a	0162		1	Daily pointer	Integer	Range 0-511
1	0	5377	1	Data block no	Integer	0~65535
	1	5378	1	State of the day	Integer	
	2	5379	1	day	Integer	In high byte
	3	5380	1	Month, year	BCD	Month in Lower byte
	4	5381	2	Working hours	LONG	
	6	5383	2	Failure hours	LONG	
	8	5385	2	Daily Total flow	LONG	Of the day
	10	5387	2	Daily Total energy	LONG	Of the day
	12	5389	2	Positive flow	LONG	Recorded at the last sec.
	14	5391	2	Negative flow	LONG	Recorded at the last sec
	16	5393	2	Positive energy	LONG	Recorded at the last sec
	18	5395	2	Negative energy	LONG	Recorded at the last sec
	20	5397	2	Tariff2	LONG	Recorded at the last sec
	22	5399	2	Tariff3	LONG	Recorded at the last sec
	24	5401	2	Max flow rate	REAL4	
	26	5403	2	Max energy rate	REAL4	
	28	5405	2	Max inlet temperature	REAL4	
	30	5407	2	Max outlet temperature	REAL4	
2	0	5409	1	Data block no	Integer	0~65535
	1	5410	1	State of the day	Integer	
	2	5411	1	day	Integer	
	...			,		
	24	5433	2	Max flow rate	REAL4	
	26	5435	2	Max energy rate	REAL4	
	28	5437	2	Max inlet temperature	REAL4	
	30	5439	2	Max outlet temperature	REAL4	
n	Data block n					
511	0	21729	1	Data block no	Integer	0~32767
	1	21730	1	State of the day	Integer	
	...			,		
	28	21757	2	Max inlet temperature	REAL4	
	30	21759	2	Max outlet temperature	REAL4	

6. MODBUS Table of Power on time

Data Blk No	address		Reg no	Variable name	format	note
n/a	0162		1	Daily pointer	Integer	Range 0-255
1	0	28929	1	Data block no	Integer	0~65535
	1	28930	1	Minute, Second	BCD	Second in low byte
	2	28931	1	Day, Hour	BCD	Hour in low byte
	3	28932	1	Year, Month	BCD	Month in low byte
2	0	28933	1	Data block no	BCD	
	1	28934	1	Minute, Second	BCD	
	2	28935	1	Day, Hour	BCD	
	3	28936	1	Year, Month	BCD	
n	Data block n					
255	0	29949	1	Data block no	BCD	
	1	29950	1	Minute, Second	BCD	
	28	29951	1	Day, Hour	BCD	
	30	29952	1	Year, Month	BCD	